

AI5707: Pattern Recognition and Machine Learning

Assignment 2: Image Reconstruction using EVD and SVD

1. Task 1: Eigenvalue Decomposition (EVD) and Singular Value Decomposition (SVD) of Square images

Convert the given colour image to a grayscale image. Represent the image as a matrix defined by:

$$\mathbf{A} \in \mathbb{R}^{n \times n}.$$

Perform the Eigenvalue Decomposition and Singular Value Decomposition.

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1},$$

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{V}^t,$$

where

- $\mathbf{Q}$ contains the eigenvectors,
- $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues.
- $\mathbf{U}$ & $\mathbf{V}$ are unitary matrices as discussed in class.
- t corresponds to transpose.

Let $\mathbf{\Lambda}_k$ denote the diagonal matrix obtained by retaining only the largest k eigenvalues and setting the remaining eigenvalues to zero.

Reconstruct the image using EVD and SVD as given below:

$$\mathbf{A}_k = \mathbf{Q}\mathbf{\Lambda}_k\mathbf{Q}^{-1}.$$

$$\mathbf{A}_k = \mathbf{U}\mathbf{\Lambda}_k\mathbf{V}^t$$

1. Reconstruct the image using at most three different values of k . Make a judicious choice for k to explain your observations.
2. For each chosen value of k , display:
 - the original image,
 - the reconstructed image,
 - the error image

$$|\mathbf{A} - \mathbf{A}_k|$$

3. Compute the reconstruction error using the Frobenius norm

$$E(k) = \|\mathbf{A} - \mathbf{A}_k\|_F.$$

4. Plot the reconstruction error $E(k)$ as a function of the number of retained components (k) – use all possible values of k here.
5. Compare the reconstruction quality obtained using EVD and SVD for the same value of k . Explain the reason for the difference.

2. Singular Value Decomposition (SVD)

Perform Singular Value Decomposition on the rectangular image.

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^t.$$

Construct the rank- k approximation

$$\mathbf{A}_k = \mathbf{U}\mathbf{\Sigma}_k\mathbf{V}^t,$$

where only the largest k singular values are retained.

Data link: <https://drive.google.com/drive/folders/1yXW3jCMkGDojqCk15fI-FRPsXqqkbHEm?usp=sharing>

1. There are two sets of images, one set is square, and the other is rectangular.
2. Perform EVD and SVD on square images.
3. Perform only SVD on the rectangular image.
4. The name of the image has the suffix for your specific group. Use only the image intended for your group.